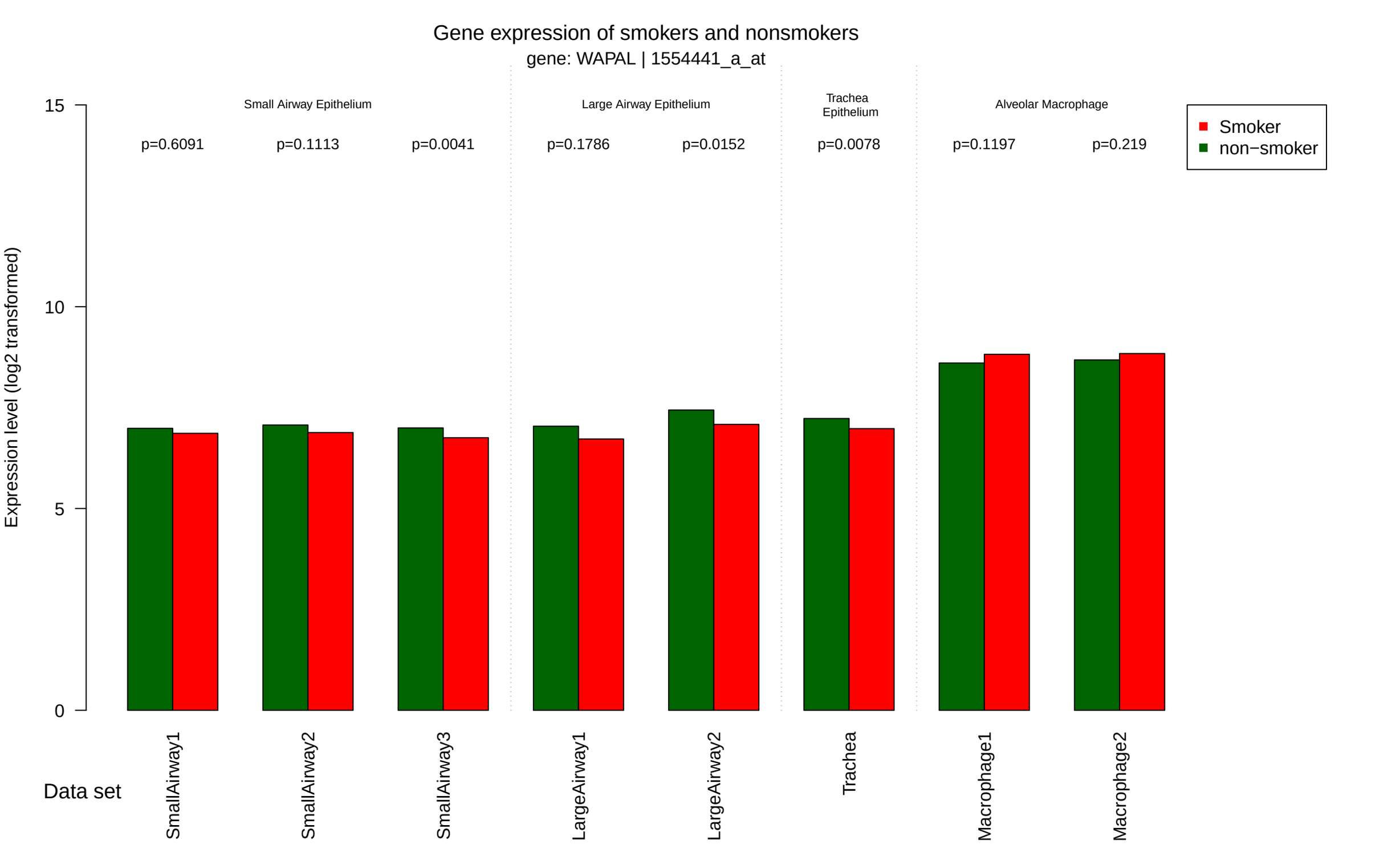
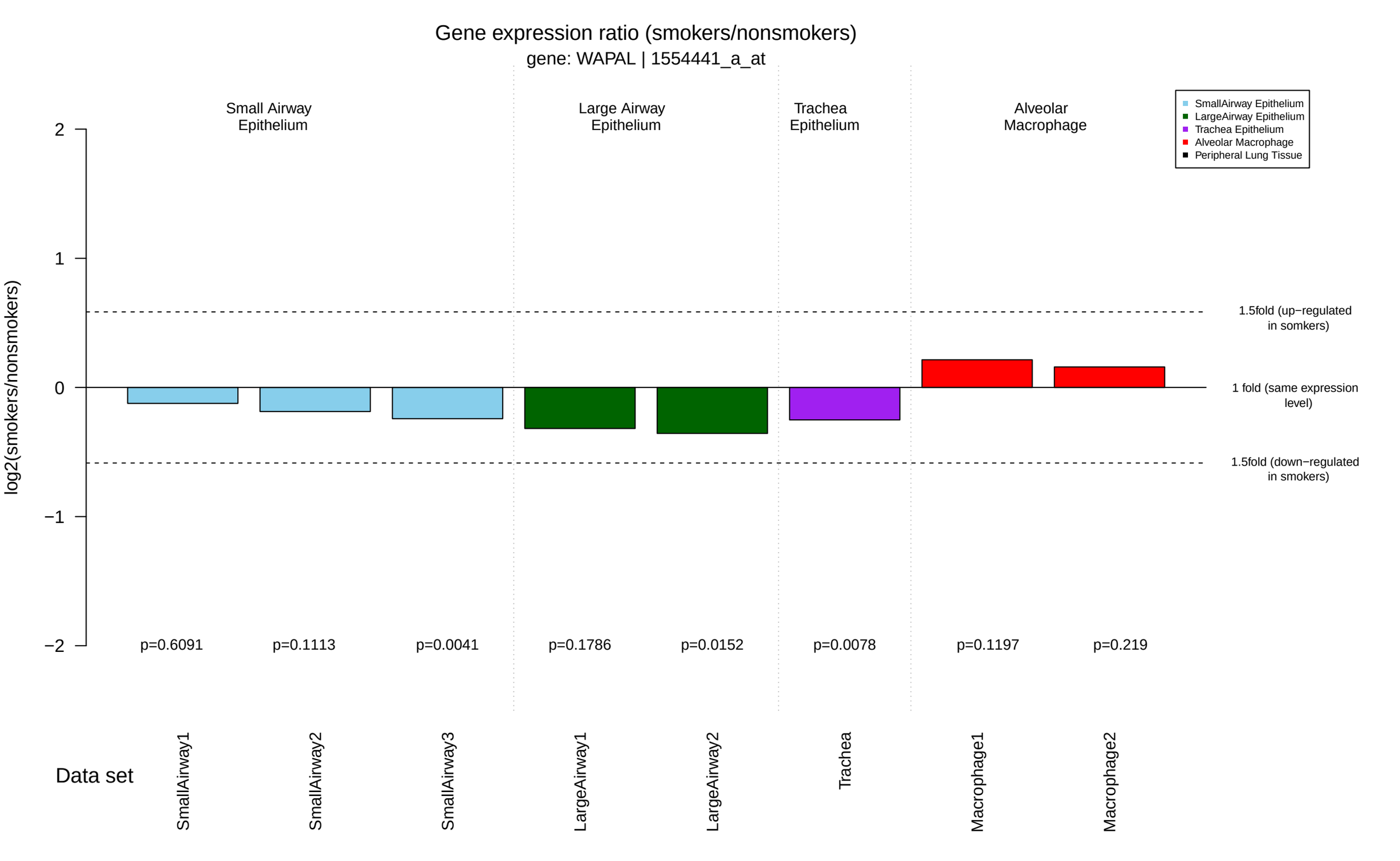
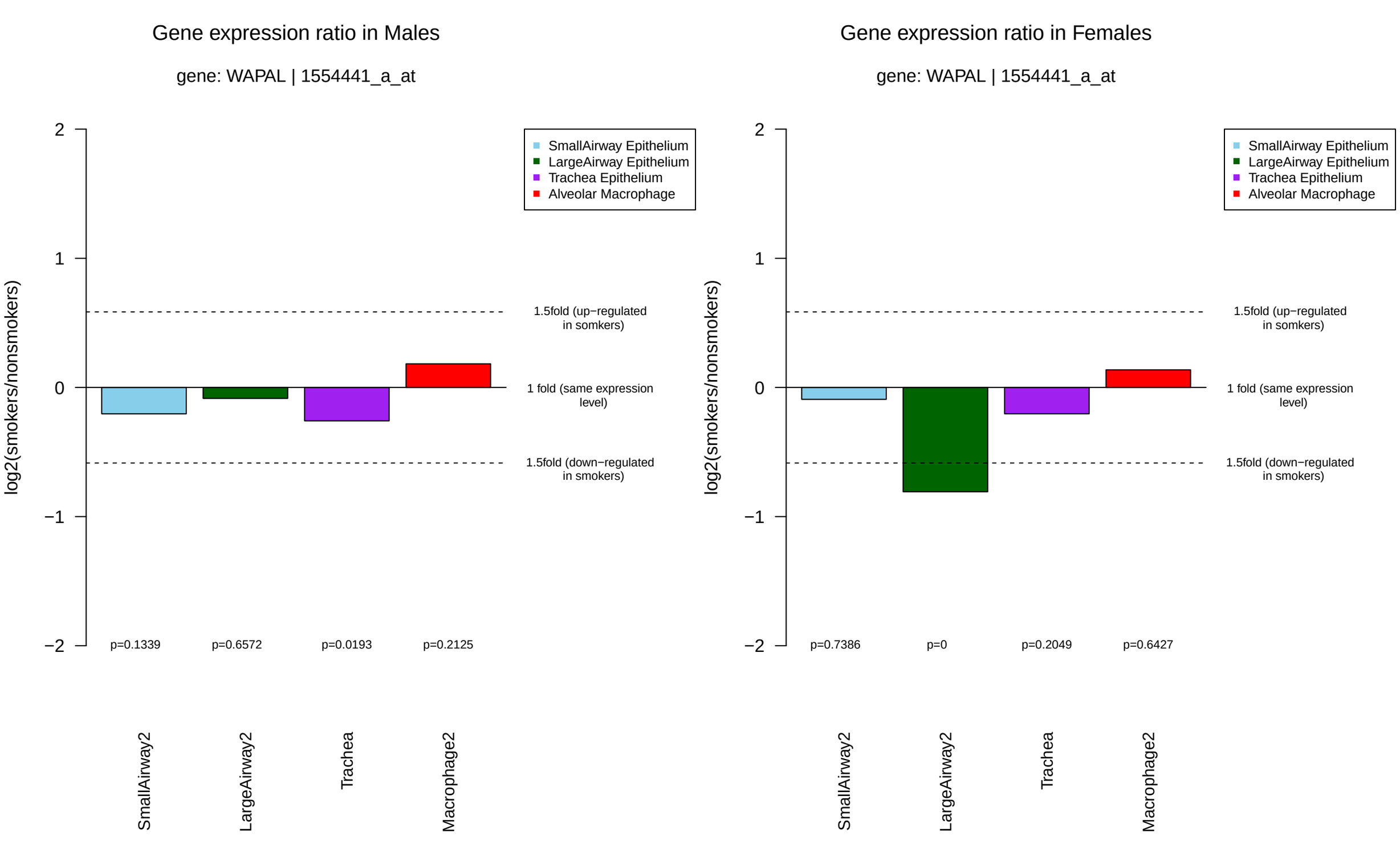
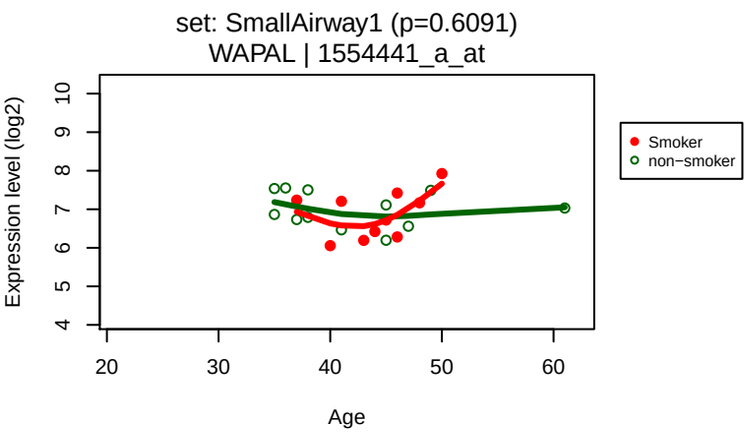
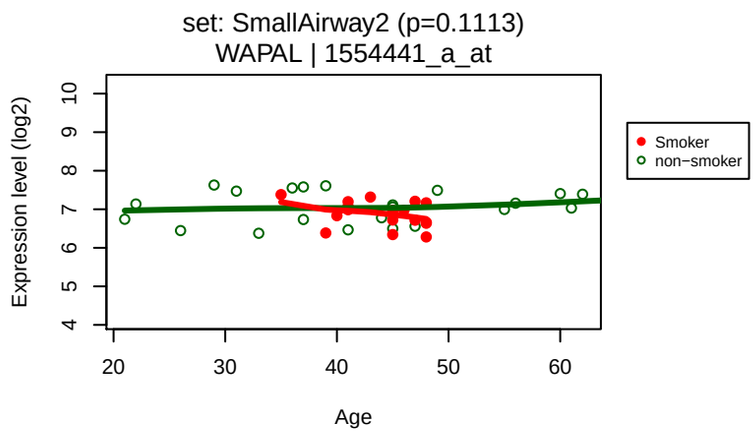
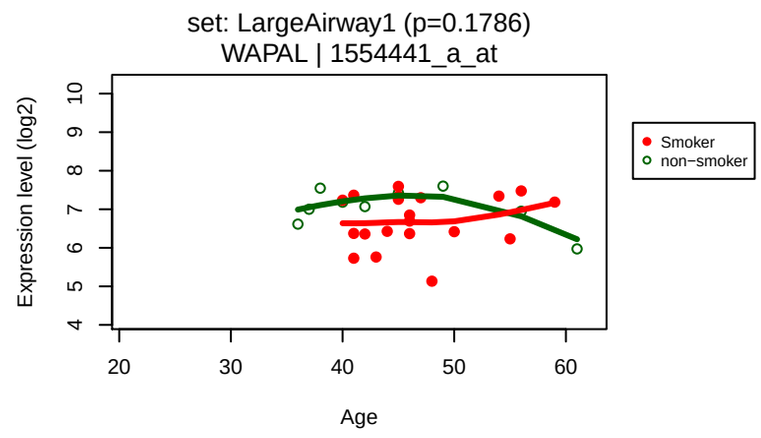

Gene expression of smokers and nonsmokers
gene: WAPAL | 1554441_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6091
p=0.1113
p=0.0041
p=0.1786
p=0.0152
p=0.0078
p=0.1197
p=0.219
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

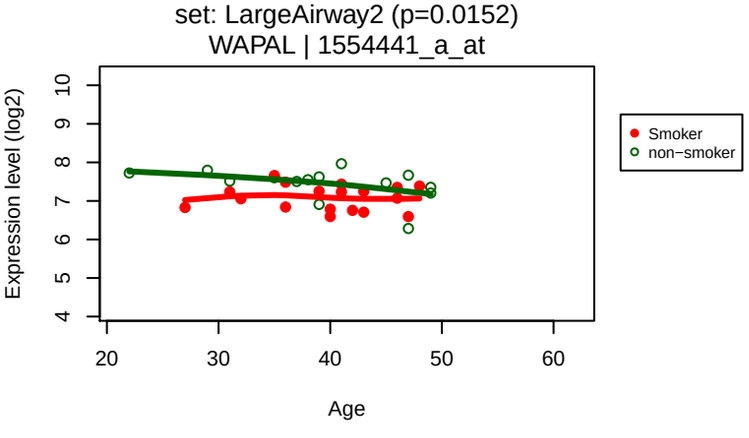

Gene expression ratio (smokers/nonsmokers)
gene: WAPAL | 1554441_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6091
p=0.1113
p=0.0041
p=0.1786
p=0.0152
p=0.0078
p=0.1197
p=0.219
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

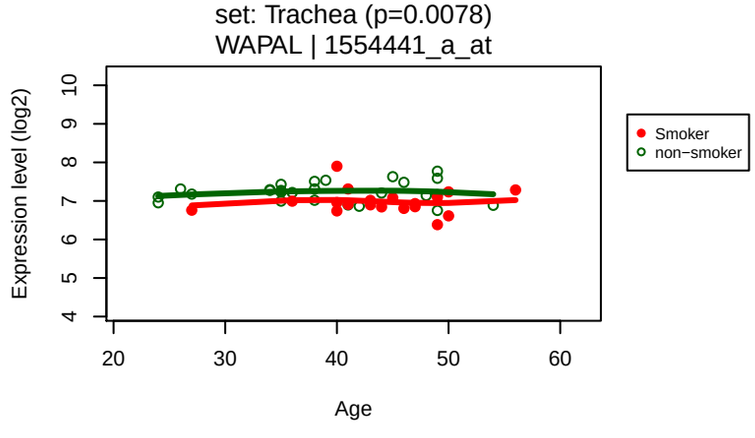

Gene expression ratio in Males
gene: WAPAL | 1554441_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1339
p=0.6572
p=0.0193
p=0.2125
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WAPAL | 1554441_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7386
p=0
p=0.2049
p=0.6427
SmallAirway2
LargeAirway2
Trachea
Macrophage2

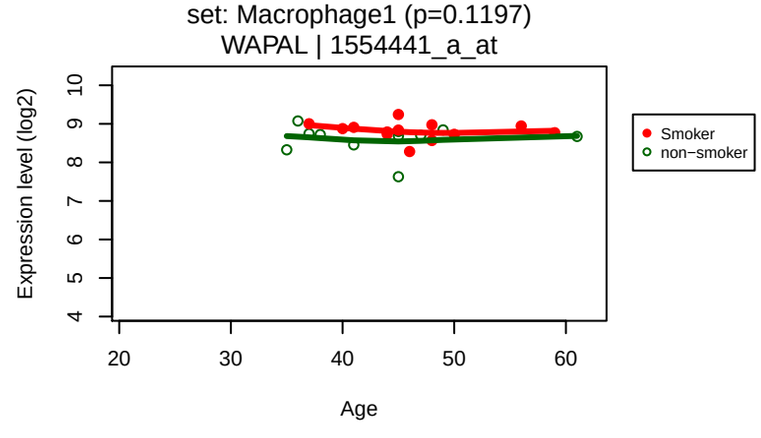

set: SmallAirway1 (p=0.6091)
WAPAL | 1554441_a_at
Expression level (log2)
Age
Smoker
non−smoker

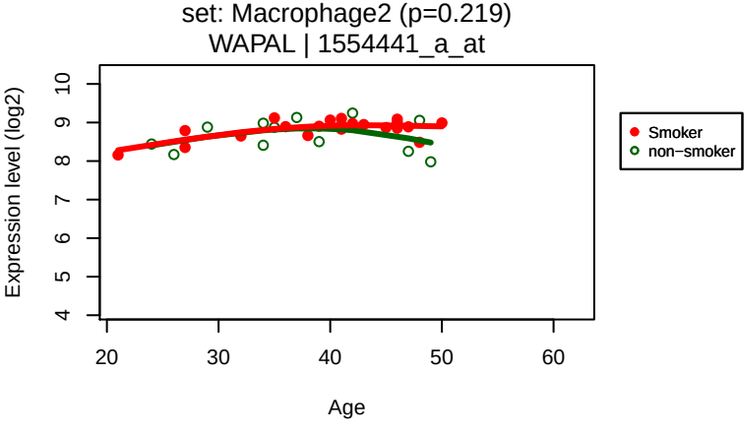

set: SmallAirway2 (p=0.1113)
WAPAL | 1554441_a_at
Expression level (log2)
Age
Smoker
non−smoker

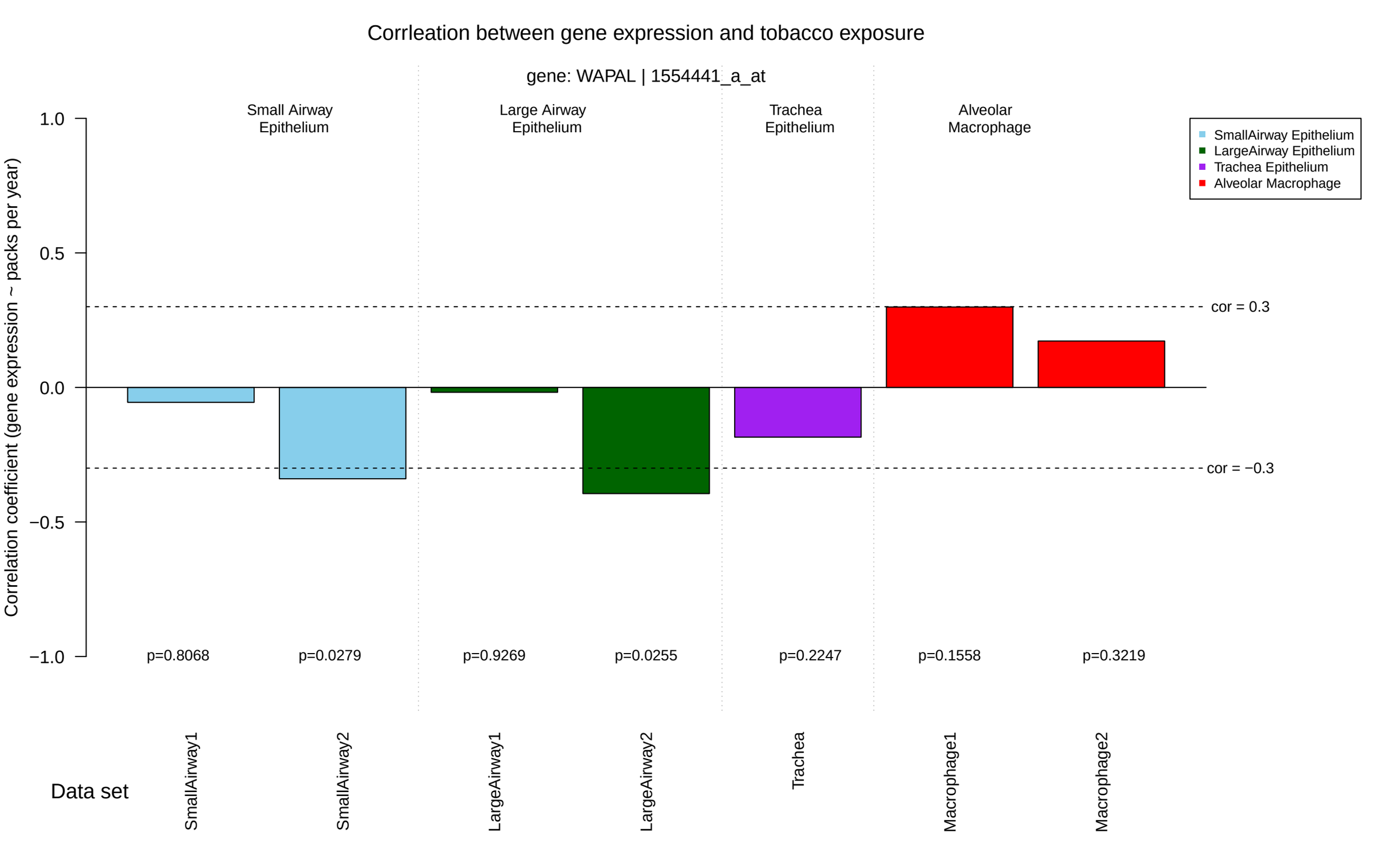

set: LargeAirway1 (p=0.1786)
WAPAL | 1554441_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0152)
WAPAL | 1554441_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0078)
WAPAL | 1554441_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1197)
WAPAL | 1554441_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.219)
WAPAL | 1554441_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WAPAL | 1554441_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8068
p=0.0279
p=0.9269
p=0.0255
p=0.2247
p=0.1558
p=0.3219
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2